

Past features must respect publication time

Daily learning note | 4 October 2026 | Q010

Sidiq review status: Not yet reviewed.

Research question and hypothesis

Can a lagged return feature use an exact session window and stay unchanged when valid future data or later revisions are added? The hypothesis is that selecting each interval's latest available version at the decision, then linking a complete fixed window, prevents those additions from rewriting the past. Missing or delayed intervals should remain unresolved rather than become zero.

Q009b separated terminal evidence from membership yesterday. Q010 is the next eligible dependency before cross-sectional ranks. It connects the existing availability selector and bounded calendar to a reusable feature calculation. All returns here are invented synthetic values. The October and November 2026 calendar labels are prospective teaching fixtures, not realised market observations.

Beginner explanation and definitions

Imagine keeping dated receipts. A receipt can describe Monday but arrive on Wednesday. A Tuesday calculation cannot use that receipt simply because its subject is Monday. A corrected receipt arriving on Thursday must not replace the version actually available on Tuesday. Return records work similarly.

Session: a listed exchange trading day, not every weekday. **Interval return:** fractional change from the preceding session's close to this session's close. **Observed time:** that ending close. **Available time:** when this version and all its underlying inputs could actually be used. **Revision:** another version of the same interval. **Latency:** extra elapsed processing time.

Periods: number of intervals included. **Skip:** most recent completed sessions deliberately omitted. **Gross factor:** 1 plus fractional return. **Linked return:** accumulated change obtained by multiplying gross factors. **Missing:** no eligible record at an exact required close; not zero. The caller explicitly names a return field; the function does not interpret price levels as returns.

Data, metrics and evaluation protocol

Use the pinned October 29-November 30 NYSE calendar, synthetic asset SYNTH and a November 4 pre-open decision. Two intervals ending November 2 and 3 have returns +0.10 and -0.10. Add a November 3 revision of +0.50 available November 5 and a November 4 return of +0.20. Compare the original feature exactly before and after extension, including its selected records; hand-check linked arithmetic to 1e-12.

Check missingness, exact publication/latency boundaries, skipped windows, asset isolation, holidays, early closes and UTC equivalence. Independently update Fraction wealth on 125 paths and compare 420 calendar/window/latency cases against explicit index and availability logic. Replay the experiment twice byte-for-byte; run full project checks. These are engineering metrics, not investment results.

Mathematics and worked example

For h included intervals with fractional returns r_1 through r_h , set $G = \text{product}(1+r_i)$ and $R = G-1$. This is geometric linking. It follows by substituting $W_i = W_{(i-1)}(1+r_i)$ repeatedly into the wealth recurrence. With no external cash flows, $W_h = W_0 \cdot G$. Returns are dimensionless; 0.10 means 10%. The inputs must describe a consistent linked wealth convention, including any reinvestment assumption.

Start at 100. A +10% interval gives 110. A subsequent -10% loses 11, leaving 99. Thus $G = 1.10 \cdot 0.90 = 0.99$ and $R = -0.01$, or -1%. Adding +10% and -10% gives zero, which is the wrong control. A -100% interval gives gross zero. Under the stipulated wealth recurrence, subsequent factors leave that wealth at zero; this does not establish that a real inactive security has valid later returns.

Scenario	Selected returns	Linked return
November 4 original decision	+0.10, -0.10	-0.01
Same decision after future additions	+0.10, -0.10	-0.01
November 5, skip one, same historical window	+0.10, +0.50	+0.65
Same later decision, one microsecond latency	+0.10, -0.10	-0.01
November 3 interval absent or delayed	+0.10, missing	Unknown
Wrong latest-vintage rule on November 4	+0.10, +0.50	+0.65, leaked

The revision arrives exactly at the November 5 decision. Zero latency permits equality; one microsecond of latency excludes it. A future extension changes no earlier eligible set. At a later decision, a newly available revision can legitimately change the same historical window.

Calendar window and information window

Let prior sessions be ordered oldest to newest. Obtain exactly $h + \text{skip} + 1$ prior sessions, including the starting close. Take the oldest $h + 1$ from that slice and use its final h closes as required interval ends. For $h=2$, $\text{skip}=0$ at November 4, start October 30 and end November 3. The November 2 interval crosses a weekend and the clock change. It still represents one exchange-session interval.

For $h=2$ at November 30, required ends are November 25 at 21:00 UTC and November 27 at 18:00 UTC. Thanksgiving is closed, November 27 closes early, and the weekend contributes no invented observations. A row stamped at 21:00 UTC on the early-close day does not match the required 18:00 close.

For each required end, retain versions satisfying $\text{available_at} \leq \text{decision_at}$ and elapsed time since availability $\geq$ latency; then take the latest publication. Unknown calendar history raises. A known calendar window with absent observations returns explicit missing timestamps and no numeric feature.

Implementation and reproduction

src/quant_research/alpha/lagged_returns.py adds lagged_return and frozen LaggedReturn. The result retains the decision, settings, start/end, all required closes, selected original observations, missing closes, gross factor and fractional value. Input order does not change selection. Asset and return field matching are exact; one asset's record cannot fill another's gap. No membership filter is applied.

Periods and skip reject booleans and non-integers. Latency must be a nonnegative timedelta. Identifiers reject whitespace and controls. Target returns must be finite built-in numbers at least -1. Duplicate versions anywhere in the supplied batch fail, including future or unrelated rows. Nonmatching close timestamps cannot fill required intervals. No rounding, forward filling or gap compression occurs.

Run make lagged-return-experiment, make check and make note NOTE_DATE=2026-10-04. The experiment script writes experiments/lagged_returns/results.json, preserving inputs and selected evidence for all cases. The experiment README describes its input obligations, results and limitations.

Observed results and validation

Check	Observed result
Future additions	Earlier full result unchanged across all 24 fixture orders
Exact Fraction wealth updates	125 three-interval paths agree
Independent index/timing oracle	420 window/latency cases agree; future revisions cannot alter them
New targeted tests	62 pass, including gaps, skips, boundaries and invalid inputs
Reproduction	Two subprocess runs match archived JSON bytes
Full project	602 Python tests; 66 C++ checks and golden output pass
Code quality	Ruff lint/47 formatted files; strict mypy/21 source files pass

Fresh warning-clean C++17 compilation, dependency consistency and four original Q003 provenance hashes also pass. An initial test incorrectly expected 20 tiny positive gross factors to underflow. Their product is still representable at 2^{-1060} ; the corrected test preserves that value even though G-1 rounds to -1. No production change was needed for that test correction. Lint then caught two missing strict zip arguments, and mypy caught reused loop-variable typing. All were corrected before full validation passed. No coverage percentage, speedup or mutation-test result is claimed.

Primary sources consulted

CFA Institute's 2020 GIPS Standards for Firms, glossary entry "link" (printed page 77), gives the geometric-linking definition. We use that arithmetic only; this project claims no GIPS compliance. https://www.gipsstandards.org/wp-content/uploads/2021/03/2020_gips_standards_firms.pdf

NYSE's official calendar confirms Thanksgiving closure, November 27 early close and ordinary core hours. The existing checksum-pinned teaching fixture supplies the bounded schedule. Accessed October 4. <https://www.nyse.com/trade/hours-calendars>

Python datetime documentation explains aware timestamps and time-zone conversion. Existing UTC normalization is reused for elapsed-time comparisons rather than local-clock subtraction. <https://docs.python.org/3/library/datetime.html>

Limitations and interview explanation

Each record asserts one complete preceding-session return with consistent identity, currency, action treatment, costs and wealth convention. Availability must include all underlying inputs and computation. A field name cannot authenticate these claims, detect backdated revisions or repair future-adjusted prices and incorrectly labelled intervals.

Q008 can provide an interval's split/dividend arithmetic. A test links its synthetic zero total return with +10% and obtains +10%. General linking still requires an explicit reinvestment convention; Q008's cash entitlement is not necessarily settled or reinvestable at the next boundary. Q009b's unresolved terminal proceeds cannot be replaced with a zero interval return or omitted to complete a window. No source adapter, cash ledger, universe selection or empirical momentum strategy is implemented.

Future-invariance assumes valid batches, unchanged original rows and honest new publication times. Malformed/duplicate future rows or a failing input generator can abort a historical query. The pinned calendar supports only its verified coverage and does not predict realised closures.

Floats are approximate. Nonfinite products and complete positive underflow raise, even if a later zero factor would restore a finite mathematical product. A tiny positive gross can coexist with a rounded value of -1; inspect `gross_return`. No exactness or measured speed is claimed.

In an interview, distinguish event and knowledge windows, derive 100-to-110-to-99, explain revision filtering and missingness, then identify unauthenticated source assumptions. Repository output does not prove personal independent mastery.

Next work and necessary input

Q010 is complete within this declared-return linking scope. Q011 cross-sectional ranks is Ready for October 5. Q014 remains today's distinct 16:00 synthesis; C003 remains Wednesday October 7. Q005b must preserve interval/basis, source and revision evidence before empirical use. No input is needed for the next synthetic task. Real acquisition still requires lawful access/source and sample evidence.

The daily schedule remains 08:00 Europe/London including daylight-saving changes. Stop new research after December 31, 2026 and request the next programme scope. Personal review remains Not yet reviewed.

Check your understanding

1. Why do +10% and -10% compound to -1%, rather than zero?
2. Can a November 5 revision of a November 3 return enter a November 4 feature?
3. What happens when a required interval is missing but older observations exist?
4. Does geometric linking of Q008 cash-entitlement returns prove cash could be reinvested?

Suggested answers

1. The loss applies to 110 after the gain, so wealth becomes 99. Multiply 1.10 by 0.90, then subtract 1.
2. No. The economic date is old enough, but its publication is too late. It can enter a later decision when both availability and processing latency are satisfied.
3. Keep the exact window and return a missing feature with the required timestamp. Do not shift the window, forward fill or substitute zero. Unknown calendar coverage instead raises an explicit error.
4. No. Arithmetic linking requires a declared wealth/reinvestment convention. Entitlement alone does not establish settlement, execution or the ability to reinvest at a boundary.